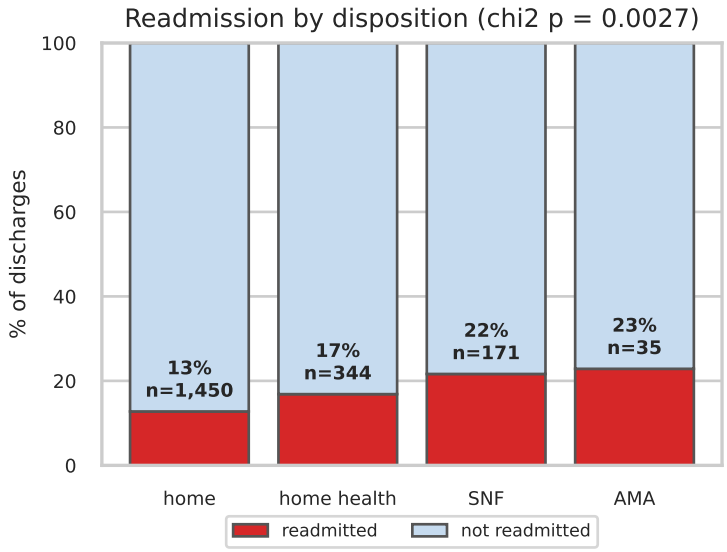
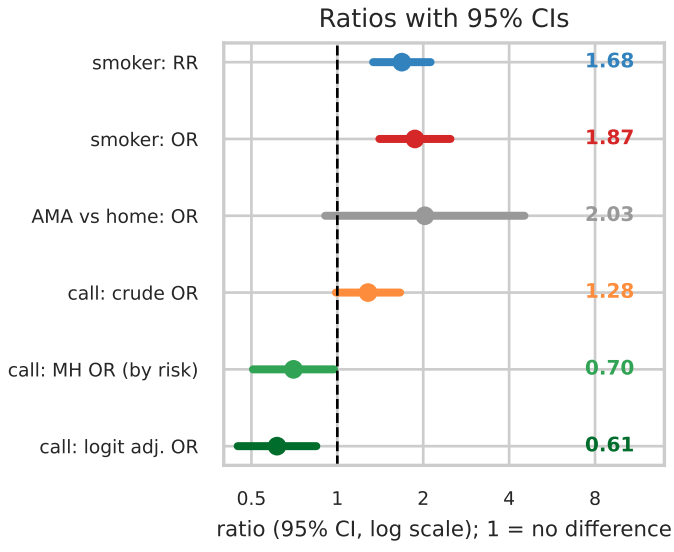
Day 8 Cheat Sheet: Categorical Outcomes

Chi-square, relative risk and odds ratios on 30-day readmissions

TERM	WHAT IT MEANS	CODE
Contingency table	counts for every combo of two categories	pd.crosstab(a, b, normalize="index")
Chi-square test	are the two variables independent?	stats.chi2_contingency(tab)
Expected count	count you'd see if there were no relationship	chi2_contingency(...)[3]
Fisher's exact	exact test for small 2x2 tables (expected < 5)	stats.fisher_exact(tab)
Cramer's V	effect size, 0 = none, 1 = perfect	sqrt(chi2 / (n * (min(r, c) - 1)))
Risk difference	risk exposed - risk unexposed (points)	a/(a+b) - c/(c+d)
Relative risk	risk exposed / risk unexposed	Table2x2(t).riskratio
Odds ratio	odds exposed / odds unexposed, odds = p/(1-p)	Table2x2(t).oddsratio
Mantel-Haenszel	one OR pooled across strata of a confounder	StratifiedTable(tables)

2x2 LAYOUT: rows = exposure (yes first), columns = outcome (readmit yes first) -> [[a, b], [c, d]]





THIS DATA (simulated): 2,000 discharges, 30-day readmission rate 14.4%
Disposition: chi2 = 14.16, dof = 3, p = 0.0027, Cramer's V = 0.084 (real but small); smallest expected count 5.04
AMA vs home: expected AMA readmits 4.5 (< 5), Fisher exact OR 2.03, p = 0.120
Smoker: 21.5% vs 12.8%, risk difference +8.7% pts, RR 1.68 (1.33 to 2.12), OR 1.87 (1.40 to 2.49)
Follow-up call: crude OR 1.28 (0.99 to 1.66, looks worse) but Mantel-Haenszel OR by risk group 0.70 (0.51 to 0.97), p = 0.0338

HABITS THAT KEEP YOU HONEST
- Report a rate with its n: "21.5% of 372 smokers", never just "21.5%"
- Check expected counts before trusting chi-square; switch to Fisher's exact when any is under 5
- A tiny p-value with a tiny Cramer's V means "real but weak"; say both
- Say "risk" for RR and "odds" for OR; when the outcome is common (over ~10%), OR > RR
- Before comparing groups, ask who got the treatment and why (stratify by the confounder)

```
tab = pd.crosstab(df["discharge_disposition"], df["readmit_30d"])
chi2, p, dof, expected = stats.chi2_contingency(tab)
tt = Table2x2(pd.crosstab(df.smoker, df.readmit_30d).loc[["yes","no"], [1,0]].to_numpy())
print(tt.riskratio, tt.riskratio_confint(), tt.oddsratio, tt.oddsratio_confint())
```

Infographic: From Counts to Odds Ratios

Simulated: 2,000 discharges, 30-day readmission (yes/no)

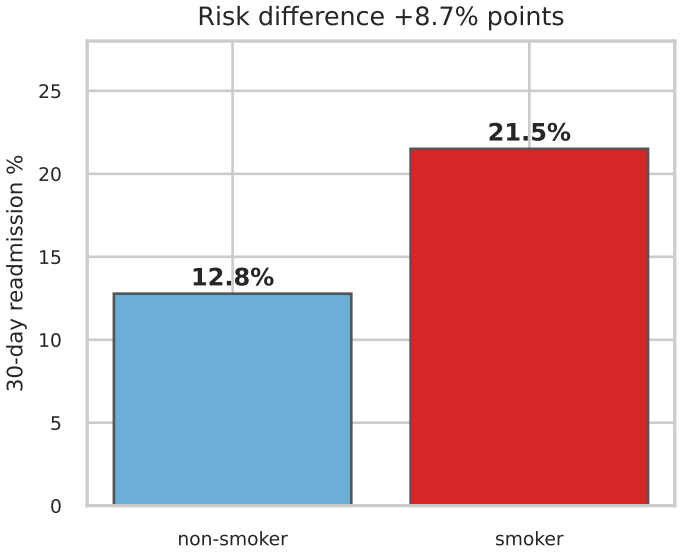
Count	Rate	Test	Size	Stratify	Model
crosstab with margins	normalize= 'index'	chi-square or Fisher exact	Cramer's V, RD, RR, OR	Mantel-Haenszel	logit: exp(coef)=OR

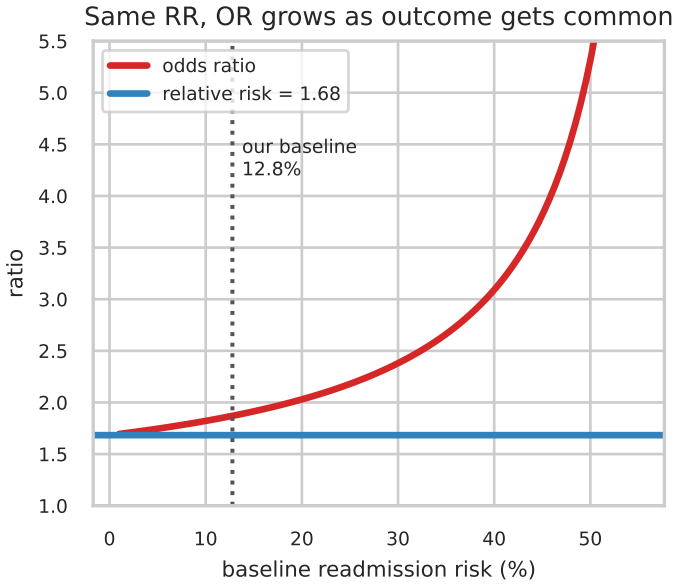
READMIT RATE
14.4%
of 2,000

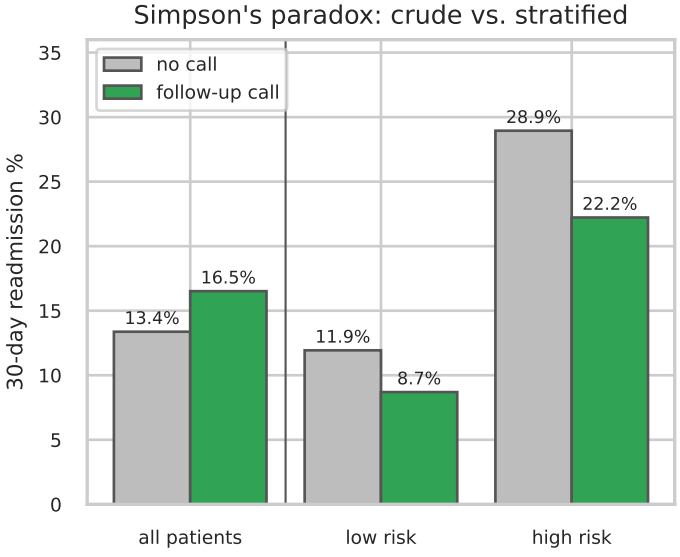
SMOKER RR
1.68
(OR 1.87)

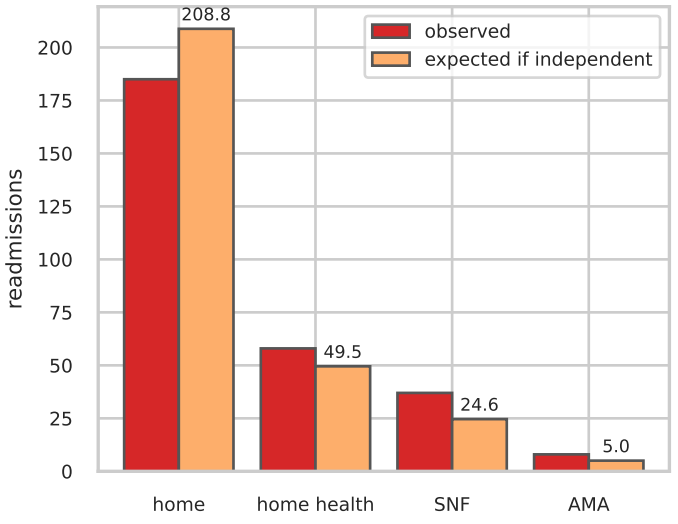
CALL, CRUDE
OR 1.28
looks worse

CALL, BY RISK
OR 0.70
helps









Why the call looked worse: 77% of high-risk patients got a call vs. 18% of low-risk patients.
Within each risk group the call group does better (8.7% vs 11.9% low risk, 22.2% vs 28.9% high risk).

Hands-On Worksheet: ds-genai-90-days / day08

60-90 minutes | notebook: day08/readmissions_categorical.ipynb

[ ] mkdir day08, copy day07/patients.csv in, conda activate ds90

[ ] Simulate readmissions.csv with seed 8 (snippet 1); rate should be 14.4%

[ ] pd.crosstab smoker x readmit_30d with margins, then normalize='index'

[ ] chi2_contingency on disposition x readmit: chi2, dof, p, expected

[ ] Smallest expected count? Compute Cramer's V and read it in words

[ ] AMA vs home 2x2: expected under 5, so run fisher_exact

[ ] Smoker 2x2 (yes first, readmit first): risk difference, RR, OR + CIs

[ ] Loop baseline risk 1% to 50%: watch the OR pull away from the RR

[ ] Follow-up call: crude rates and crude OR. Does the call 'hurt'?

[ ] Stratify by risk_group; StratifiedTable for the Mantel-Haenszel OR

[ ] smf.logit: check exp(coef) for smoker equals the 2x2 OR (1.87)

[ ] Findings cell; git add day08 && git commit && git push

[ ] Stretch: Snowflake COUNT_IF readmission rate table by disposition

```
import numpy as np, pandas as pd
from scipy import stats
from statsmodels.stats.contingency_tables import Table2x2, StratifiedTable
df = pd.read_csv("readmissions.csv")
tab = pd.crosstab(df["discharge_disposition"], df["readmit_30d"])
chi2, p, dof, expected = stats.chi2_contingency(tab)       # expect chi2 ~14.16, p ~0.0027
t2 = pd.crosstab(df["smoker"], df["readmit_30d"]).loc[["yes", "no"], [1, 0]]
tt = Table2x2(t2.to_numpy())
print(tt.riskratio, tt.oddsratio)                          # expect ~1.68 and ~1.87
```

Readmit rate: _____ chi2: _____ p: _____ Cramer's V: _____ Fisher p: _____

Smoker RD: _____ RR: _____ OR: _____ Call crude OR: _____ MH OR: _____

Day 8 Self-Check Quiz

1. What does the 'expected count' in a chi-square test represent?

_____

2. AMA has an expected readmission count of 4.5. Which test do you use, and why?

_____

3. Chi-square p = 0.0027 but Cramer's V = 0.084. How do you describe this result?

_____

4. Smokers: 21.5% readmitted vs. 12.8%. Give the risk difference and RR in plain words.

_____

5. Why is the odds ratio (1.87) bigger than the relative risk (1.68)?

_____

6. Calls look worse overall (OR 1.28) but helpful in each risk group. What happened?

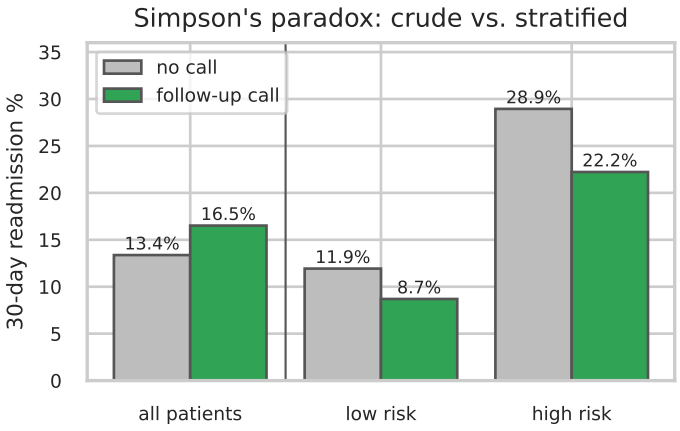
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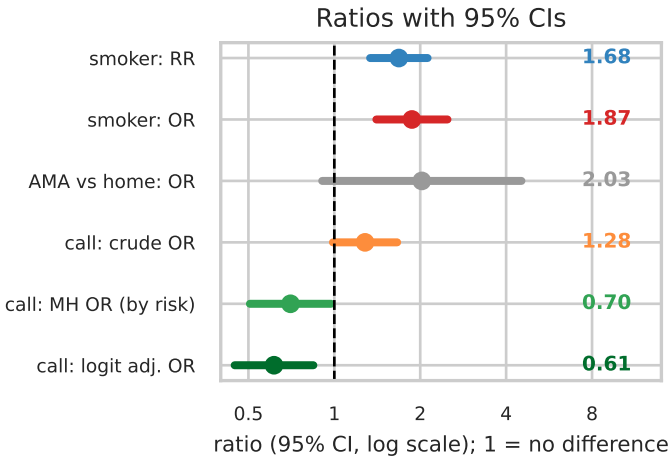
7. What does the Mantel-Haenszel OR of 0.70 tell a care-management manager?

_____

8. In smf.logit, how do you turn a coefficient into something a manager can read?

_____





Answers:
1 the count you'd get in each cell if the two variables were unrelated: row total x column total / grand total
2 Fisher's exact test (expected < 5 makes chi-square unreliable): OR 2.03, p = 0.120, so not significant with only 35 AMA patients
3 real but weak: disposition is linked to readmission, yet it explains only a small part of who comes back
4 RD: 8.7 extra readmissions per 100 smoker discharges; RR: smokers are 1.68 times as likely to be readmitted (95% CI 1.33 to 2.12)
5 odds = p/(1-p) inflate as p grows; with a 14% outcome the OR runs above the RR. Only for rare outcomes is OR close to RR
6 Simpson's paradox: calls went mostly to high-risk patients (77% vs 18%), and high risk drives readmission. Risk group is a confounder
7 among patients of the same risk group, a call goes with about 30% lower odds of readmission (95% CI 0.51 to 0.97); observational, not proof
8 exponentiate it: np.exp(coef) is the odds ratio, holding the other predictors fixed (smoker: 1.87, same as the table)